Case Study 4: Water saving in Golf courses

	User net benefit (USD)	User net benefit (USD) per tonne of biochar	User cost (USD)	Payback	Years of effects
Golf course water saving[v] Average golf course	$2,055,000	$22,600	$200,000	< 1 year	10 yr.
Golf course water saving Capital lease – average golf course	$1,674,000	$18,500	0	0 years	10 yr.

Colorado based company *Biochar Now* first began selling biochar to golf courses in California[vi] five years ago[vii], according to CEO James Gaspard.

Infestations of beetle in Colorado forests have produced voluminous amounts of unused wood biomass which is the feedstock of the biochar product. *Biochar Now*'s biochar making process focuses on producing high temperature, high carbon chars using modernised slow pyrolysis ring kilns over 8-10 hours, contrasting with other regional producers who focus on fast pyrolysis. Emissions (PM2.5 , NO_x, CO) from these kilns are very low.

The company's pyrolysis units do not capture oils or gases, focusing on high C content biochar. The product was recently tested by a potential client, and the *Biochar Now* product exhibited 20% higher carbon than its nearest competitor.

US$10 million has been invested to get *Biochar Now* to its current point, with US$1 million in sales per annum now being achieved. James estimates a minimum biochar sale price of $1 per pound ($2.20 / kg), indicating a maximum current annual production of 450 metric tonnes, though prices can be as high as $3 per pound ($6.61 / kg) depending on the grade of the biochar product and the scale of the order.

James stated that the results and performance of biochar in Californian golf courses can only be associated with the 'Biochar Now' biochar.

Southern California has sandy coarse textured soils, in which biochar has been shown to retain soil moisture[21]. It is not uncommon in the Southern Californian summers to experience 110°F (43°C) days in popular residential and tourist destinations such as Coachella valley and Palm Springs which is home to 150 golf courses. These golf courses can consume 1 million gallons of water per day at an annual expense of between $0.5 – $1 million in water irrigation bills. Climate change and increasing severity of droughts is expected to exacerbate these costs.

Biochar Now biochar is applied raw at a rate of approximately 2% at the root zone. Initially, there were some dust challenges associated with biochar handling. To solve this, the company developed a pelletized product. The golf course (with biochar) is then watered at normal rates for a week, then watering is cut back thereafter. The biochar holds moisture to six times its dry weight, and the biochar

[v] Assuming highest cost / lowest effect (the least financially feasible scenario) of 30% water irrigation cost savings and high cost biochar application for average sized golf course of $200,000. The lowest cost / highest effect (65% irrigation savings cost and $100,000 application cost) results in net benefits greater by an order of magnitude, but the lower bound has been considered only to maintain conservative estimates of user value. The 10 year effect is also a conservative estimate, with biochar expected to continue delivering cost savings for the duration of its physical persistence, which is effectively in perpetuity. Assuming ongoing effects doubles the estimate of financial value.

[vi] This US case study is outside of the Australia & New Zealand scope, but an exception was made to include it to demonstrate a novel biochar use that has many applications in both Australia and New Zealand.

[vii] Five years prior to March 2019.

is able to reduce both water and fertiliser use by 50%. 1.6% of biochar in root zone doubles turf grass growth.

The raw char sells to golf courses at a rate of $2000 - $6000 per metric tonne of char. To apply biochar to these golf courses using *Biochar Now's* raw product costs $100k - $200k. While expensive, this reduces irrigation bills by $300k - $500k with a payback of approximately 6 months.

Initially the upfront cost of the biochar was a barrier for clients. The solution was a financing arrangement (capital lease), where golf courses amended with biochar paid the standard amount of water expense, and any water cost saving went to the financial institution over one to two years. After this period, the arrangement ended, and the golf course received the ongoing savings. This arrangement was able to show 30-40% interest (IRR), and so was of great appeal to financiers. The char lasts indefinitely (half-life of 17,000 years according to lab trials), with consistent effects over time without issues of erosion or char translocation within the soil profile.

Despite the focus on carbon purity, revenues from carbon credits remain elusive. James estimates that it would cost $0.5 million to do the requisite paperwork to claim $0.75 million in revenues. At current carbon prices, the company would be lucky to obtain $10 per tonne CO_2e in the voluntary market.

However, the methodology framework for biochar is improving rapidly, with researchers seeking to address concerns relating to modelling nitrous oxide soil emissions. Within the next few years it is anticipated that issuance of these credits will become simpler for biochar producers.

Other lines of business:

Biochar Now has tested in 13 major markets, including for improved hemp production and other specialty agriculture, applications in building materials (asphalt, plastics, cement), environmental remediation and algal bloom reduction. *Biochar Now* is targeting 'bottomless markets'.

James estimates that they have $100 million backlog in orders from the hemp production industry, where he estimates his larger grade product can increase revenues by $100,000 per hectare due to improved crop yields. In light of this, customers happily outlay $2,000-$4,000 per hectare for biochar.